

# **Impact of Generative Artificial Intelligence on Software Engineering Jobs**

## **Executive Summary**

Generative Artificial Intelligence (GenAI) is poised to significantly impact the workforce, with potential productivity gains and trillions of dollars in value across sectors [S1]. Breakthroughs in foundation models have enabled new capabilities and vastly improved existing ones across various modalities, including images, video, audio, and computer code [S1]. GenAI has the potential to automate 60 to 70 percent of employees' time today, changing the anatomy of work and augmenting individual workers by automating some of their activities [S1]. The technology could add \$2.6 trillion to \$4.4 trillion annually across 63 use cases, increasing the impact of all artificial intelligence by 15 to 40 percent [S1]. GenAI will have a significant impact across all industry sectors, with banking, high tech, and life sciences seeing the biggest impact as a percentage of their revenues [S1]. The value of GenAI use cases falls across four areas: customer operations, marketing and sales, software engineering, and R&D; with 75 percent of the value delivered across these areas [S1].

## **Generative Ai Adoption And Productivity Gains**

Generative AI has undergone significant advancements in the last decade, transitioning from rule-based systems to sophisticated models generating human-like text, images, and software code [S3]. GenAI can fill gaps and accelerate solutions within the software development lifecycle, collaborating with human software engineers in activities such as inception and planning, code generation, code translation and migration, test case generation, documentation, and architecture and design analysis [S3]. A study by Ness Digital Engineering and Zinnov found that GenAI implementation increases productivity by 70%, reduces task completion time for existing code updates by 70%, and task completion time for senior engineers by 48% [S2]. Additionally, GenAI enables a ~10% reduction in high code complexity tasks and improves engagement by 70% [S2]. The study's findings suggest that GenAI complements rather than dictates workflows, facilitating frictionless knowledge sharing and unlocking the true value of globalization [S2].

GenAI can increase coding productivity by enhancing efficiency, reducing manual effort, and freeing up developers to focus on creative activities, with potential productivity improvements of 20-40% in code quality for Greenfield projects [S3]. GenAI also facilitates software design productivity through early research analysis, rapid prototypes, enhanced collaboration, virtual design, and design automation, although evidence is less conclusive than with coding and test tasks [S3]. Furthermore, GenAI tools and technologies are being integrated into software development workflows to enhance team collaborations, with various studies indicating productivity improvements [S3].

## **Job Market Implications And Organizational Readiness**

Generative AI has emerged as a key field of study due to its flexible applications, with organizations leveraging it for diverse applications, including IT, cybersecurity, code generation, writing, and marketing content creation [S4]. The preliminary findings of a study reveal that 97% of IT workers use Generative AI tools, mainly ChatGPT, resulting in significant personal productivity gains and perceived organizational efficiency improvements that correlate positively with Generative AI adoption [S4]. However, increased organizational adoption of AI strongly correlates with heightened employee job security concerns [S4]. Key adoption challenges include inaccurate outputs (64.2%), regulatory compliance issues (58.2%), and ethical concerns (52.2%) [S4]. Studies have shown that GenAI can automate code generation, enhance firm-level productivity, and drive economic growth, but its economic returns depend heavily on organizational readiness and process integration [S4].

Organizational readiness is crucial for the successful adoption of GenAI, as it requires significant changes in workflows, processes, and employee skills [S4]. The findings of a study offer early empirical insights into GenAI's economic and organizational implications, highlighting the need for targeted research in this area [S4]. As GenAI continues to evolve and improve, it is essential for organizations to assess their readiness and develop strategies to mitigate the potential risks and maximize the benefits of GenAI adoption [S4].

## **Conclusion**

Generative Artificial Intelligence (GenAI) has the potential to significantly impact the workforce, with potential productivity gains and trillions of dollars in value across sectors [S1]. GenAI can fill gaps and accelerate solutions within the software development lifecycle, collaborating with human software engineers in various activities [S3]. The technology can

increase coding productivity, facilitate software design productivity, and enhance team collaborations, with various studies indicating productivity improvements [S3]. However, organizational readiness is crucial for the successful adoption of GenAI, as it requires significant changes in workflows, processes, and employee skills [S4]. As GenAI continues to evolve and improve, it is essential for organizations to assess their readiness and develop strategies to mitigate the potential risks and maximize the benefits of GenAI adoption [S4]. The impact of GenAI on software engineering jobs will depend on the ability of organizations to adapt and integrate this technology into their workflows [S4].

## References

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